



## OR - operations research

Operations Research (Technische Universität München)



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**Esolution**

Place student sticker here

**Note:**

- During the attendance check a sticker containing a unique code will be put on this exam.
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## Operations Research

**Exam:** IN0024 / Endterm  
**Examiner:** Prof. Dr. Martin Bichler

**Date:** Friday 2<sup>nd</sup> August, 2024  
**Time:** 08:30 – 10:30

|    | P 1 | P 2 | P 3 | P 4 | P 5 | P 6 | P 7 |
|----|-----|-----|-----|-----|-----|-----|-----|
| I  |     |     |     |     |     |     |     |
| II |     |     |     |     |     |     |     |

### Working instructions

- This exam consists of **20 pages** with a total of **7 problems**.  
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 120 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **non-programmable pocket calculator**
  - one **analog dictionary** English ↔ native language
  - one **ruler/geo triangle**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Answer the exam questions in English or German.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from \_\_\_\_\_ to \_\_\_\_\_ / Early submission at \_\_\_\_\_

## Problem 1 Multiple-Choice Questions (14 credits)

Mark all correct answers (there can be more than one):

Mark correct answers with a cross

To undo a cross, completely fill out the answer option

To re-mark an option, use a human-readable marking



a) Given a linear program, degeneracy:

- ☒ can be recognized if at least one basic variable has value 0.
- ☐ means that the linear program has multiple optimal solutions.
- ☐ means that the Simplex algorithm will not terminate.
- ☐ implies that the linear program is unbounded.

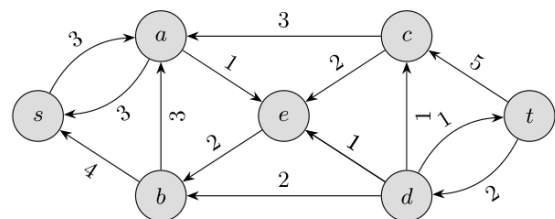
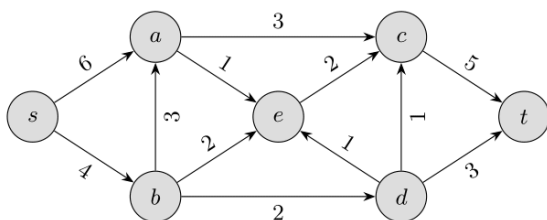
b) Given a linear program (P) and its dual (D):

- ☐ If (P) is infeasible, then (D) has to be unbounded.
- ☐ If (P) is infeasible, then (D) has to be infeasible.
- ☐ If (P) is unbounded, then (D) has to be unbounded.
- ☒ If (P) is unbounded, then (D) has to be infeasible.

c) Which of the following statements is/are true regarding column generation (CG)?

- ☒ If the pricing problem does not produce a solution with positive objective value, then CG terminates.
- ☐ If the pricing problem does not produce a solution with positive objective value, then there are errors in the formulation of the pricing problem.
- ☐ CG eliminates more and more constraints after every iteration.
- ☐ CG is particularly useful when the linear programming problem has a small number of variables but a large number of constraints.

d) Consider a network  $N$  with source  $s$  and sink  $t$  and its residual network corresponding to a maximum flow (see figures below). The numbers on the edges indicate their capacities.



Select all minimum  $s - t$  cuts in the network  $N$ :

- ☐  $\{s, a, b\}, \{c, d, e, t\}$
- ☐  $\{s, a\}, \{b, c, d, e, t\}$
- ☒  $\{s, a, b, c, e\}, \{d, t\}$
- ☒  $\{s, a, b, e\}, \{c, d, t\}$

e) Why are total unimodularity of the constraint matrix  $A$  and having an integral constraint vector  $b$  important for solving network flow problems?

- ☒ They ensure that the solution to the linear programming relaxation is an integer solution.
- ☐ They reduce the sizes of the network flow problems, making them more manageable.
- ☐ They guarantee that the Simplex algorithm will find the optimal solution in polynomial time.
- ☐ They provide a heuristic approach to approximate solutions.

f) Which of the following statements is/are true?

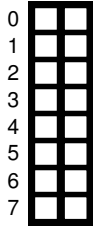
- ☒ The capacity of a cut in a network separating source and sink is always greater or equal to the maximum flow.
- ☐ The maximum flow in a network is equal to the sum of all capacities of the network.
- ☐ The capacity of a cut in a network separating source and sink is always smaller or equal to the maximum flow.
- ☒ The maximum flow in a network is equal to the capacity of the minimum cut separating the source and the sink.

g) Which of the following functions is/are convex?

- ☐  $f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = x^T A x, \forall A \in \mathbb{R}_{>0}^{n \times n} (A = (a_{ij})_{i,j \in \{1, \dots, n\}}, a_{ij} > 0).$
- ☐  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \exp(\cos(x)).$
- ☒  $f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \|x\|, \text{ for an arbitrary norm } \|\cdot\|.$
- ☒  $f, f_i : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = -\min_{i \in I} f_i(x), \text{ where } f_i \text{ is concave and } I \text{ is a finite index-set.}$

Additional space for notes:

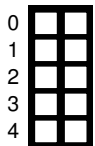
## Problem 2 Duality (16 credits)



a) Derive the dual linear program from the following linear program (P).

$$\begin{aligned}
 (P) \quad & \text{Max} \quad 2x_1 - x_2 + 3x_3 \\
 & \text{s.t.} \quad x_1 - 2x_2 + 2x_3 \leq 4 \\
 & \quad \quad x_1 + x_2 + x_3 \leq 8 \\
 & \quad \quad -x_1 + x_2 - 3x_3 \geq -4 \\
 & \quad \quad x_3 \geq 0
 \end{aligned}$$

$$\begin{aligned}
 (D) \quad & \text{Min} \quad 4y_1 + 8y_2 - 4y_3 \\
 & \text{s.t.} \quad y_1 + y_2 - y_3 = 2 \\
 & \quad \quad -2y_1 + y_2 + y_3 = -1 \\
 & \quad \quad 2y_1 + y_2 - 3y_3 \geq 3 \\
 & \quad \quad y_1, y_2 \geq 0 \\
 & \quad \quad y_3 \leq 0
 \end{aligned}$$



b) State all complementary slackness conditions.

Primal CS conditions:

$$\begin{aligned}
 (x_1 - 2x_2 + 2x_3 - 4)y_1 &= 0 \\
 (x_1 + x_2 + x_3 - 8)y_2 &= 0 \\
 (-x_1 + x_2 - 3x_3 + 4)y_3 &= 0
 \end{aligned}$$

Dual CS conditions:

$$\begin{aligned}
 (y_1 + y_2 - y_3 - 2)x_1 &= 0 \\
 (-2y_1 + y_2 + y_3 + 1)x_2 &= 0 \\
 (2y_1 + y_2 - 3y_3 - 3)x_3 &= 0
 \end{aligned}$$

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c) Use the complementary slackness conditions to check whether  $x^T = (6, 2, 0)$  is an optimal solution for  $(P)$ .

First, we check the feasibility of  $x$  by inserting it into  $(P)$  and obtain

$$\begin{aligned}x_1 - 2x_2 + 2x_3 &= 6 - 4 && \leq 4 \\x_1 + x_2 + x_3 &= 6 + 2 && \leq 8 \\-x_1 + x_2 - 3x_3 &= -6 + 2 && \geq -4 \\x_3 &= 0 && \geq 0,\end{aligned}$$

so that  $x$  satisfies all primal constraints. It is thus a feasible point in  $(P)$ .

Since the first primal constraint is not tight, from the first primal CS condition it follows that  $y_1 = 0$ . Furthermore, as  $x_1$  and  $x_2$  are strictly positive, from the dual CS conditions, the following must hold

$$\begin{aligned}y_1 + y_2 - y_3 - 2 &= 0 \\-2y_1 + y_2 + y_3 + 1 &= 0.\end{aligned}$$

After substituting the value of  $y_1 = 0$ , we get

$$\begin{aligned}y_2 - y_3 &= 2 \\y_2 + y_3 &= -1.\end{aligned}$$

We therefore get  $y^T = (0, 1/2, -3/2)$ .

Finally we check the dual feasibility of the determined point

$$\begin{aligned}y_1 + y_2 - y_3 &= \frac{1}{2} - \left(-\frac{3}{2}\right) && = 2 \\-2y_1 + y_2 + y_3 &= \frac{1}{2} - \frac{3}{2} && = -1 \\2y_1 + y_2 - 3y_3 &= \frac{1}{2} - 3\left(-\frac{3}{2}\right) && \geq 3 \\(y_1, y_2) &= (0, 1/2) && \geq 0 \\y_3 &= -3/2 && \leq 0.\end{aligned}$$

So  $y$  is feasible for  $(D)$ .

It follows from the above argument that  $x$  must be an optimal solution for  $(P)$ .

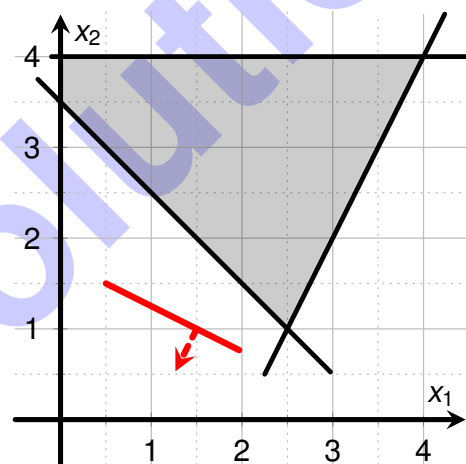
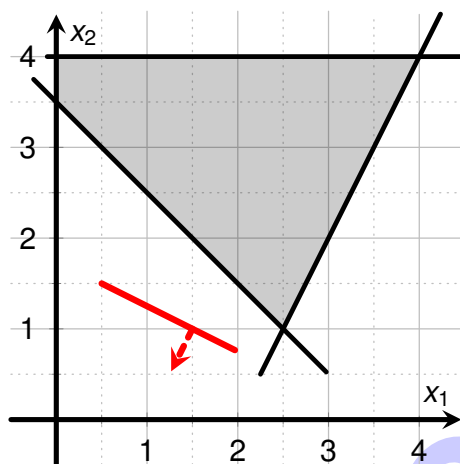
### Problem 3 Branch-and-Bound (16 credits)

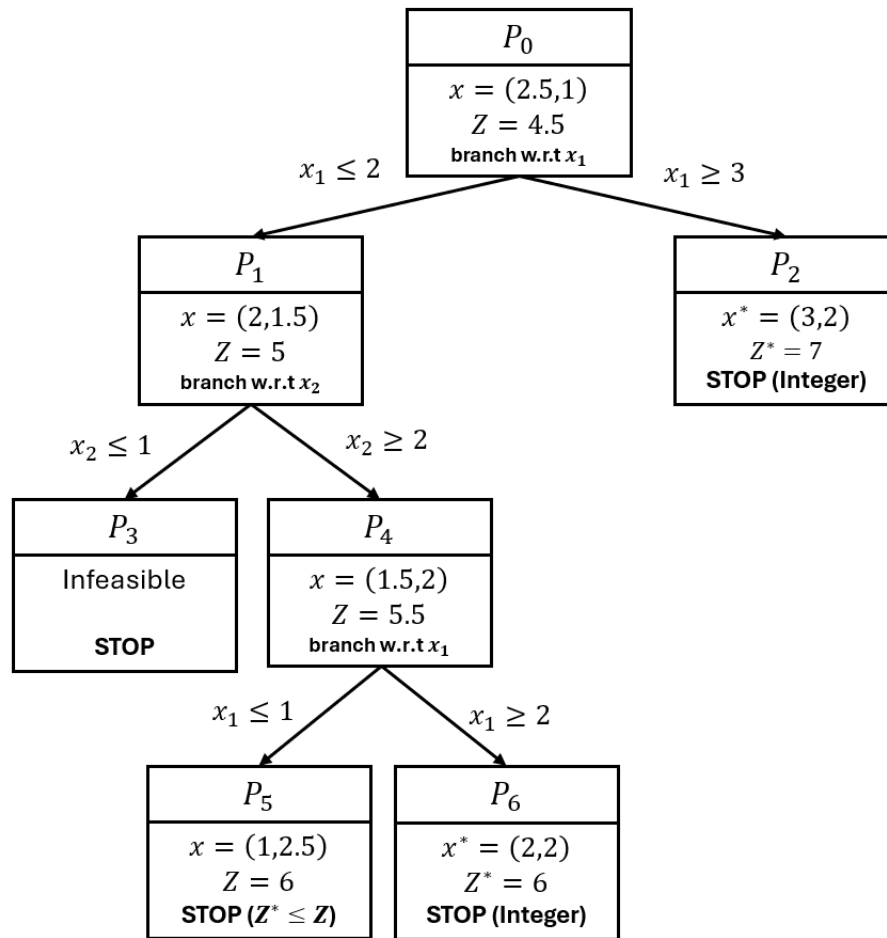
Consider the following minimization problem, given as an integer linear program (P)

$$\begin{aligned} (P) \quad & \text{Min} \quad x_1 + 2x_2 \\ & \text{s.t.} \quad 2x_1 + 2x_2 \geq 7 \\ & \quad \quad -2x_1 + x_2 \geq -4 \\ & \quad \quad -x_2 \geq -4 \\ & \quad \quad x_1, x_2 \geq 0 \\ & \quad \quad x_1, x_2 \in \mathbb{Z} \end{aligned}$$

Solve the integer linear program (P) using the branch-and-bound method. Specify the optimal values for  $x_1$  and  $x_2$ , as well as the optimal value of the objective function.

Note: In the following coordinate systems, the feasible set of the LP relaxation of (P) and the objective function are shown. You can use them to solve the subproblems graphically.





An optimal solution of the problem (P) is given by  $x^* = (2, 2)$  with objective value  $Z^* = 6$ .

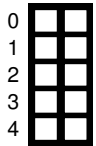


## Problem 4 IP-Modelling - Tour d'Allemagne (22 credits)

As the *Tour de France* gains in popularity, the politician Freddy Merckz wants to propose a similar professional cycling event in Germany, the *Tour d'Allemagne*. He asks you to create a tour that adheres to the constraints given by the organizers.

A tour consists of 21 consecutive stages. In each stage  $j \in J = \{1, \dots, 21\}$ , the athletes have to complete a route  $i \in I$ . There are 50 possible routes, i.e.,  $I = \{1, \dots, 50\}$ , from which you can choose for the various stages.

Formulate an integer linear program to help the organizers designing a tour. Use the binary variable  $x_{i,j} \in \{0, 1\}$  with  $i \in I, j \in J$ , where  $x_{i,j} = 1$  if route  $i$  is selected for stage  $j$  and  $x_{i,j} = 0$  else.



a) As hosting a route brings prestige to a town, each town (uniquely associated to a route) pays  $p_i \geq 0$  to the organizers if their route  $i$  is part of the tour. They even pay double if their route  $i$  is chosen for the first or last stage. Formulate an objective function to determine the most profitable tour for the organizers.

$$\begin{array}{ll} \text{Max} & \sum_{i \in I} p_i \cdot \left( 2 \cdot x_{i,1} + \sum_{j=2}^{20} x_{i,j} + 2 \cdot x_{i,21} \right) \\ \text{s.t.} & \end{array}$$



Formulate the following constraints as linear (in-)equalities. You may introduce additional variables. If you do so, please also state their intuitive meaning in words:

b) Each stage consists of exactly one route.

$$\sum_{i \in I} x_{i,j} = 1, \quad \forall j \in J$$



c) Each route can be selected at most once for the whole tour.

$$\sum_{j \in J} x_{i,j} \leq 1, \quad \forall i \in I$$

d) Let the parameter  $d_{i,i'} \geq 0$  denote the distance between route  $i$  and  $i'$  in km. To ensure that the equipment can be transported in time, the distance between routes of consecutive stages has to be less than 200 km.

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We introduce a binary variable  $y_{i,i'} \in \{0, 1\}$ : route  $i'$  comes in the stage right after the stage with route  $i \Rightarrow y_{i,i'} = 1$ :

$$x_{i,j} + x_{i',j+1} \leq 1 + y_{i,i'}, \quad \forall i, i' \in I, j \in J \setminus \{21\}$$

Then, we can formulate the constraint on the distance:

$$y_{i,i'} d_{i,i'} \leq 200, \quad \forall i, i' \in I$$

Sample Solution

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e) The parameter  $m_i$  describes if a route  $i$  is classified as a mountain route ( $m_i = 1$ ) or not ( $m_i = 0$ ). There have to be more than 3 mountain routes in either the first or the last 7 stages (not in both).

Note: We assume that the set of possible routes contains enough mountain routes, such that a feasible solution exists.

We introduce a binary variable  $z \in \{0, 1\}$ : more than 3 mountain routes in first 7 stages  $\Leftrightarrow z = 1$ :

$$\sum_{j=1}^7 \sum_{i \in I} m_i x_{i,j} \leq 3 + Mz$$

$$\sum_{j=1}^7 \sum_{i \in I} m_i x_{i,j} \geq 4z$$

Using this variable, we can now formulate the rest of the constraint:

$$\sum_{j=15}^{21} \sum_{i \in I} m_i x_{i,j} \leq 3 + M(1 - z)$$

$$\sum_{j=15}^{21} \sum_{i \in I} m_i x_{i,j} \geq 4(1 - z)$$

For  $M$  we can choose any fixed number greater or equal 4.

Note that one could also introduce two binary variables. One for the first ( $z_1$ ) and one for the last ( $z_2$ ) seven stages, similar to  $z$ . The XOR constraint could then be written as  $z_1 + z_2 = 1$ .

Sample Solution

## Problem 5 Matroids (14 credits)

Let  $T = (V, E)$  be a tree with nodes  $V$  and edges  $E$ .

a) Consider two distinct nodes  $s, t \in V$ ,  $s \neq t$ . Let  $U_1 = (E, \mathcal{I}_1)$ , where

$$\mathcal{I}_1 = \{F \subseteq E : F \text{ is a subset of edges of an } s - t \text{ path in } T\}.$$

Prove that  $U_1$  is a matroid.

We prove that  $U_1$  is a matroid.

- $\emptyset \in \mathcal{I}_1$ , since a subset of edges can be empty.
- Let  $A \in \mathcal{I}_1$  and  $B \subseteq A$ . Then it holds that  $B \in \mathcal{I}_1$ .
- Let  $A, B \in \mathcal{I}_1$  such that  $|A| < |B|$ . Since in a tree the path between two nodes is unique, there exists an edge  $e \in B \setminus A$  such that  $A \cup \{e\} \in \mathcal{I}_1$ .

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b) A subset  $S \subseteq V$  is called *stable* if for all nodes  $u, v \in S$  it holds that  $\{u, v\} \notin E$ . Let  $U_2 = (V, \mathcal{I}_2)$ , where  $\mathcal{I}_2 = \{S \subseteq V : S \text{ is stable in } T\}$ . Provide an example which shows that  $U_2$  is not a matroid.

Consider the tree  $T = (V, E)$  with  $V = \{a, b, c\}$  and  $E = \{\{a, b\}, \{a, c\}\}$ . The subsets  $A = \{a\}$  and  $B = \{b, c\}$  are stable and it holds that  $|A| < |B|$ . But neither  $\{a, b\}$  nor  $\{a, c\}$  is stable and thus there exists no  $v \in B \setminus A$  such that  $A \cup \{v\} \in \mathcal{I}_2$ .

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Sample Solution

## Problem 6 Tom and the ATMs (18 credits)

Tom is the mayor of Frosthaven and has received many complaints on the same topic: there is no ATM in Frosthaven. The town is represented as an undirected graph  $G = (V, E)$ , where  $E$  is the set of streets and  $V$  denotes the set of endpoints of the streets. Tom wants to solve an optimization problem to find the minimum number of ATMs that must be placed in Frosthaven, such that for each street  $(i, j) \in E$  there is an ATM located at  $i$  or at  $j$  (or at both).

- 0  a) Formulate an integer program that encodes Tom's optimization problem. Use the variable  $x_i \in \{0, 1\}$  with  $i \in V$ , where  $x_i = 1$  if an ATM is placed at point  $i$ .

$$\begin{aligned} \min \quad & \sum_{i \in V} x_i \\ \text{s.t.} \quad & x_i + x_j \geq 1 \quad \forall (i, j) \in E \\ & x_i \in \{0, 1\} \quad \forall i \in V \end{aligned}$$

- 0  b) To which NP-hard problem learned in the lecture does Tom's optimization problem correspond (no justification needed)?

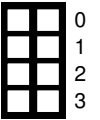
Vertex cover

- 0  c) How many ATMs must be placed in Frosthaven if  $G$  was a complete graph with  $|V| = n$  (no justification needed)?

$n - 1$

An optimal number of ATMs have been placed in Frosthaven. Now, Tom wants to visit all ATMs and check them. Let  $U \subseteq V$  be the set of points where ATMs have been placed. Assume that between any points  $i, j \in U, i \neq j$  there exists a street  $\{i, j\} \in E$  and  $l_{ij} \in \mathbb{R}_+$  denotes its length. Tom wants to compute a tour of minimum length which starts at a point  $i \in U$ , visits all points  $j \in U$  with  $j \neq i$  exactly once, and then returns to  $i$ . The tour should not include points in  $V \setminus U$ .

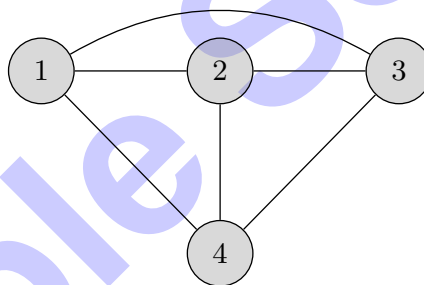
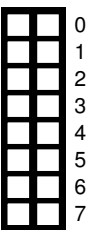
d) To approximately solve this problem, Tom identified 2 approximation algorithms,  $ALG_1$  and  $ALG_2$ .  $ALG_1$  has an approximation ratio of 2, while  $ALG_2$  has an approximation ratio of 1.5. He says "I will use  $ALG_2$  to find an approximate solution. Since  $ALG_2$  has a better approximation ratio compared to  $ALG_1$ , it must yield a better result!". Is Tom's statement correct? Justify your answer.



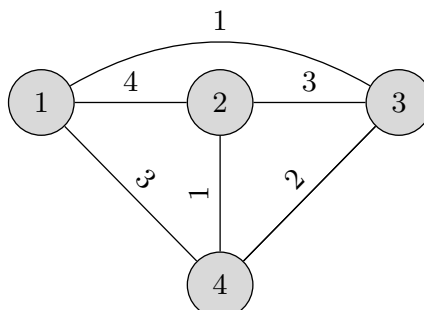
No. An algorithm with an approximation ratio of  $p$  guarantees that, in the worst case, the solution it provides is no more than  $p$ -times worse than the optimal solution. However, it is not guaranteed that  $ALG_2$  will compute a better solution than  $ALG_1$ .

e) Tom knows that one street of length  $K$  connecting two points in  $U$  is under maintenance and is thus not navigable. He identified an optimal tour  $T$  of length  $L$  where the length of any street connecting two consecutive points in  $T$  is strictly smaller than  $K$ . He claims that there exists no other tour of length  $L$  that would include the street under maintenance. Provide an example which shows that Tom's claim is generally false. Explain your example.

Hint: Use the graph below and assume that node 1 is the starting point, and its edge to node 2 is the one under maintenance.



Consider the following example:

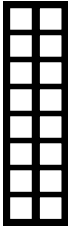


Edge  $\{1, 2\}$  is the street under maintenance, thus  $K = 4$ . Therefore, any tour without edge  $\{1, 2\}$  starting at node 1 has to be of the form  $1 - x - 2 - y - 1$ . Therefore, there are exactly two tours ( $1 - 4 - 2 - 3 - 1$  and  $1 - 3 - 2 - 4 - 1$ ) with identical edges, both with minimal length 8. For each edge  $\{i, j\}$  in, for example,  $1 - 4 - 2 - 3 - 1$ , it holds that  $l_{ij} < K$ . Another optimal tour with length 8 that contains the street under maintenance is  $1 - 2 - 4 - 3 - 1$ .



## Problem 7 Non-Linear Optimization (20 credits)

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a) Find all local extrema of the following function

$$f(x, y) = x^2 + y^2 + e^{2x} - (8x + 6y) + 2xy$$

and determine their type (max/min).

$$\nabla f(x, y) = \begin{pmatrix} 2x - 8 + 2y + 2e^{2x} \\ 2y - 6 + 2x \end{pmatrix}$$
$$\nabla^2 f(x, y) = \begin{pmatrix} 2 + 4e^{2x} & 2 \\ 2 & 2 \end{pmatrix}$$

Since  $\text{Tr}(\nabla^2 f(x, y)) > 0$  and  $\det(\nabla^2 f(x, y)) > 0$  it follows that  $\nabla^2 f(x, y)$  is positive definite. Alternative: principal minor criterion. Hence,  $f$  is strictly convex. Consequently, if a minimum exists it is unique. No maxima exist. We investigate the critical points of  $f$  ( $\nabla f = 0$ ). We solve the corresponding system of equations

$$\begin{aligned} x + y + e^{2x} &= 4 \\ x &= 3 - y \end{aligned}$$

by substituting the second equation into the first one and solving for  $y$ , we obtain

$$y = 3, x = 0$$

The unique extreme point is thus  $(x^*, y^*) = (0, 3)$ , which is a global minimum.

b) Find all local minima of  $f$  under the additional constraint

$$g_1(x, y) = x + y - 2 \leq 0$$

We want to find extrema of a convex function on a convex set. Since  $g_i$  is convex and there exists a strictly feasible point, it follows that slaters condition holds. Therefore, the KKT-conditions are sufficient and necessary for optimality. We define the lagrangian for minimization as

$$L_-(x, y, \lambda_1, \lambda_2) = f + \lambda_1 g_1$$

The gradient of the constraint is

$$\nabla g_1(x, y) = (1, 1)^T$$

For the critical points of the lagrangian we obtain

$$\begin{pmatrix} 2x - 8 + 2y + 2e^{2x} \\ 2y - 6 + 2x \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

We distinguish two cases:

1.  $g_1$  is tight
  2.  $g_1$  is not tight
1. Substituting  $x = 2 - y$  yields

$$\begin{pmatrix} 2(2 - y) - 8 + 2y + 2e^{2(2-y)} \\ 2y - 6 + 2(2 - y) \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

Which reduces to

$$\begin{pmatrix} -4 + 2e^{2(2-y)} \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

By dual feasibility it must hold that  $\lambda_1 \geq 0$ . Hence, we obtain  $\lambda_1 = 2 > 0$  and consequentially

$$e^{2(2-y)} = 1 \Leftrightarrow y = 2$$

Thus, we found a first minimizer  $(x^*, y^*) = (0, 2)$

2.  $x + y < 2$ . By complementary slackness it follows that  $\lambda_1 = 0$ . Therefore we need to solve

$$\nabla f(x, y) = \begin{pmatrix} 2x - 8 + 2y + 2e^{2x} \\ 2y - 6 + 2x \end{pmatrix} = 0$$

The second equation is equivalent to  $x + y = 3$  which contradicts  $x + y < 2$ .

Alternatively: The unique solution computed in a) is unfeasible.

Hence, no additional KKT-points. In summary  $(0, 2)$  is the unique local/global minimizer of  $f$ .

c) Let  $f : K \rightarrow \mathbb{R}$  be a strictly convex, continuously differentiable function and let  $K \subset \mathbb{R}^n$  be convex. If  $f$  has a local minimum on  $K$ , it is unique. Give a counterexample where the local minimum of  $f$  is not unique if  $K$  is non-convex.

$\min x^2$  s.t.  $x^2 \geq 1$ . Global/local minima are  $x_{1,2} = \pm 1$ .

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution



Sample Solution